From sheets to a database

Data engineering · Level 3 · Lesson 1



Today we will

- Say what a database adds to a spreadsheet.
- Store each fact once, in tables linked by keys.
- Read a CREATE TABLE and run a first SELECT.

Remember

- One shared copy, with types and rules.
- Store each fact once.
- Primary keys pick out rows; foreign keys point to them.

One fact, one place

1. Warm-up (5 min): How many copies of one phone number?
2. Teach (8 min): Tables and keys.
3. Split (15 min): Two tables, one key.
4. Change (7 min): A new phone number.
5. Wrap-up (5 min): The same tables in SQL.

Use made-up names and phone numbers in the wide table. Make at least one class have several pupils, so the repeats are easy to see.

Let’s discuss

1. What went wrong when we changed the phone in the wide table?
2. Which column links the pupils table to the classes table?
3. Why is a phone number stored as text?

Exit ticket

- Name a primary key and a foreign key in our two tables, and say what each one does.